

Cause Level	Description	Failure Category	Evidence/Source
	anomalous software behavior.		

SDLC Analysis

Root Cause / Defect	SDLC Phase of Origin	SDLC Phase of Detection	Cost of Late Detection
Missing Brake Override System	Requirements	Operations (Fatal crashes)	Catastrophic. Required fleet-wide software/hardware updates and caused fatal accidents.
Spaghetti Code / Task Death	Design & Architecture	Operations / Legal Discovery	Catastrophic. Survived internal testing and NASA audit; only caught via legal source-code discovery. Resulted in billions in fines.
Lack of ECC RAM	Architecture	Operations	Hardware limitation meant single-bit flips could corrupt global variables undetected.
Ignoring field failure data	Maintenance / Monitoring	Operations	Resulted in criminal fraud charges for the organization.